2D Matrices: Rotations

Questions 1–27

Question 1. Describe the following linear transformations:

a) $\begin{pmatrix} 0 & -1 \\ 1 & 0 \end{pmatrix}$

b) $\begin{pmatrix} -1 & 0 \\ 0 & -1 \end{pmatrix}$

c) $\begin{pmatrix} 0 & 1 \\ -1 & 0 \end{pmatrix}$

Question 2. The matrix $\mathbf{A}$ represents a rotation by 90° anticlockwise about the origin.

a) Write down the matrix $\mathbf{A}$.

b) Find $\mathbf{A}^2$.

c) Find $\mathbf{A}^3$.

d) Find $\mathbf{A}^4$.

e) Find $\mathbf{A}^5$.

f) Find $\mathbf{A}^{2025}$.

Question 3. A shape with area 4 is mapped to its image by the matrix $\begin{pmatrix} 0 & -1 \\ 1 & 0 \end{pmatrix}$. Write down the area of the shape's image.

Question 4. The point $A(6, -2)$ is rotated by $90°$ anticlockwise about the origin to the point B. Find the coordinates of the point B.

Question 5. The point $A(\sqrt{2}, -2\sqrt{2})$ is rotated by 180° about the origin to the point B. Find the coordinates of the point B.

Question 6. The point $A(5, -17)$ is rotated by 990° about the origin to the point B. Find the coordinates of the point B.

Question 7. The triangle T has vertices at $(1, 3)$, $(2, 8)$ and $(6, -1)$. T is rotated by $270°$ anticlockwise about the origin to form T'. Find the coordinates of the vertices of T'.

Question 8. The rectangle R has vertices at $(0, 2)$, $(0, 6)$, $(10, 6)$ and $(10, 2)$. R is rotated by $180°$ about the origin to form R'. Find the coordinates of the vertices of R'.

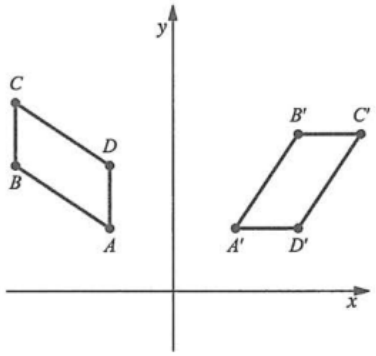
Question 9. The points A and B have coordinates $(2, -5)$ and $(5, 2)$. The point A is mapped to the point B by a rotation about the origin.

a) Fully describe the transformation.

b) Write down the matrix associated with the transformation you described in a).

c) A particle at point A travels to point B by moving around the circumference of a circle. Find the exact distance travelled by the particle from point A to point B.

Question 10. The vertices of a parallelogram $ABCD$ have coordinates $A(-2, 2)$, $B(-5, 4)$, $C(-5, 6)$ and $D(-2, 4)$. The parallelogram is mapped to the parallelogram $A'B'C'D'$, as shown in the diagram below, by the rotation about the origin associated with the matrix $\mathbf{M}$. The coordinates of D' are $(4, 2)$. Find the matrix $\mathbf{M}$ and the coordinates of A', B' and C'.



Question 11. The point $A(9, 4)$ is rotated by $90°$ anticlockwise about the point $B(6, -3)$. Find the coordinates of the image of the point A.

Question 12. Write down the matrix that represents the following rotation about the origin:

a) $45°$, anticlockwise.
b) $45°$, clockwise.
c) $135°$, anticlockwise.
d) $135°$, clockwise.

Question 13. The triangle T has vertices at $(-2, 5)$, $(8, 3)$ and $(4, -2)$. T is rotated by 45° anticlockwise about the origin to form T'. Find the exact coordinates of the vertices of T'.

Question 14. The rectangle R has vertices at $(-3, 2)$, $(-3, 4)$, $(6, 4)$ and $(6, 2)$. R is rotated by 225° clockwise about the origin to form R'. Find the exact coordinates of the vertices of R'.

Question 15. The triangle X has vertices at $(-\sqrt{2}, 3\sqrt{2})$, $(k\sqrt{2}, 5\sqrt{2})$ and $(1, -1)$, where k is a positive constant. X is rotated by 135° anticlockwise about the origin to form X', which has vertices $(-2, -4)$, $(-13, 3)$ and $(0, \sqrt{2})$ respectively. Find the value of k.

Question 16. Write down the matrix that represents the following linear transformations:

a) A rotation by 30°, anticlockwise about the origin.
b) A rotation by 60°, anticlockwise about the origin.
c) A rotation by 120°, anticlockwise about the origin.
d) A rotation by 150°, anticlockwise about the origin.
e) A rotation by 210°, clockwise about the origin.

Question 17. Write down the matrix that represents a rotation by 18°, anticlockwise about the origin, giving each of its elements to 3 significant figures.

Question 18. Write down the matrix that represents a rotation by 25°, clockwise about the origin, giving each of its elements to 3 significant figures.

Question 19. Each of the following matrices represents a rotation. Describe fully the single transformations given by:

a) $\begin{pmatrix} -\frac{1}{2} & -\frac{\sqrt{3}}{2} \\ \frac{\sqrt{3}}{2} & -\frac{1}{2} \end{pmatrix}$

b) $\begin{pmatrix} -\frac{\sqrt{3}}{2} & \frac{1}{2} \\ -\frac{1}{2} & -\frac{\sqrt{3}}{2} \end{pmatrix}$

c) $\begin{pmatrix} -\frac{\sqrt{2}}{2} & -\frac{\sqrt{2}}{2} \\ \frac{\sqrt{2}}{2} & -\frac{\sqrt{2}}{2} \end{pmatrix}$

d) $\begin{pmatrix} \frac{\sqrt{10+2\sqrt{5}}}{4} & \frac{1-\sqrt{5}}{4} \\ \frac{-1+\sqrt{5}}{4} & \frac{\sqrt{10+2\sqrt{5}}}{4} \end{pmatrix}$

Question 20. The matrix $\begin{pmatrix} \frac{5}{\sqrt{41}} & -\frac{4}{\sqrt{41}} \\ \frac{4}{\sqrt{41}} & \frac{5}{\sqrt{41}} \end{pmatrix}$ represents a rotation by θ degrees, anticlockwise about the origin. Find the value of θ to 3 significant figures.

Question 21. The matrix $\begin{pmatrix} \frac{\sqrt{6}-\sqrt{2}}{4} & \frac{\sqrt{6}+\sqrt{2}}{4} \\ -\frac{\sqrt{6}+\sqrt{2}}{4} & \frac{\sqrt{6}-\sqrt{2}}{4} \end{pmatrix}$ represents a rotation by θ degrees, anticlockwise about the origin. Find the value of θ.

Question 22. Determine which of the following matrices represent rotations:

a) $\begin{pmatrix} \frac{13}{85} & -\frac{84}{85} \\ \frac{84}{85} & \frac{13}{85} \end{pmatrix}$

b) $\begin{pmatrix} -\frac{13}{85} & -\frac{84}{85} \\ \frac{84}{85} & \frac{13}{85} \end{pmatrix}$

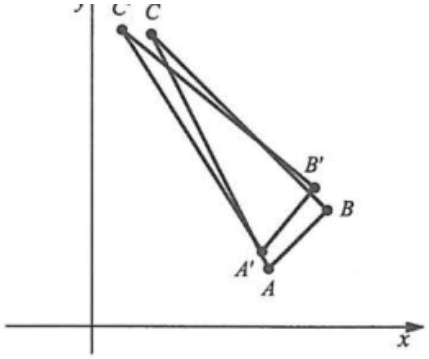
c) $\begin{pmatrix} \frac{13}{85} & \frac{84}{85} \\ -\frac{84}{85} & \frac{13}{85} \end{pmatrix}$

d) $\begin{pmatrix} \frac{84}{85} & -\frac{13}{85} \\ \frac{13}{85} & \frac{84}{85} \end{pmatrix}$

Question 23. The point $A(2, 10)$ is rotated by $\theta°$ anticlockwise about the origin to the point $B(-10, -2)$. Find the value of θ to 3 significant figures.

Question 24. The point $A(15, -2)$ is rotated by $\theta°$ anticlockwise about the origin to the point $B(2, -15)$. Find the value of θ to 3 significant figures.

Question 25. The points $A(3,1)$, $B(4,2)$, and $C(1,5)$ form a triangle. Triangle ABC is rotated about the origin to the triangle $A'B'C'$. The rotation is associated with the matrix $\mathbf{M}$. The coordinates of C' are $\left(\frac{1}{2}, \frac{\sqrt{103}}{2}\right)$. Find $\mathbf{M}$ and the coordinates of A' and B' to 3 decimal places.



Question 26. The point $A(-5,1)$ is rotated by $60°$ anticlockwise about the point $B(2,0)$. Find the coordinates of the image of the point A.

Question 27. The point $A(0, 8)$ is rotated by $30°$ anticlockwise about the point P, onto its image $A'(-5+\sqrt{3}, 3+3\sqrt{3})$. Find the coordinates of P.